

ISE 555: Advanced Deterministic Systems Optimization — Traceability

September 14 and 16, 2026

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Folder-level traceability

1 Delivery scope — iteration 007

All nine images in `raw/notes/14_09_2026` form ONE study document and ONE traceability companion, each compiled from one LaTeX source. Dates are internal sections, not separate deliverables. The user explicitly replaced the daily split from iteration 006 with this folder-level rule. The original photos remain unchanged. The date-specific evidence below is retained as internal organization within this single companion; references to daily delivery in historical descriptions are superseded.

September 14, 2026 — provenance and review

2 Scope and provenance

Course: ISE 555, Advanced Deterministic Systems Optimization. Lecture date: September 14, 2026, from the dates written on the source pages. Source folder: `raw/notes/14_09_2026/`. The September 14 part covers IMG_7831 through IMG_7834. The September 16 part covers the remaining images in the same folder.

The full initial transcription is retained in `work/2026-09-14/transcription.md`. The initial academic review and subsequent confirmations remain in that batch's working directory. The study PDF integrates explanatory material directly; this companion records its provenance without interrupting the study text.

3 Section-to-source mapping

- Operations research workflow: IMG_7831:B02, five phases of OR.
- Limitations of elementary calculus: IMG_7831:B03 and IMG_7832:B01, including endpoints, local optima, higher derivatives, numerical accuracy and discrete variables.
- Optimization categories and formulation: IMG_7832:B02–B03, deterministic systems, six formulation/solution steps and algorithm categories.
- Two-variable LP: IMG_7833:B01, objective, inequalities, triangle, vertices and values.
- Cabinet manufacturing: IMG_7833:B02 and IMG_7834:B01–B02, product table, resource constraints, general LP notation and recorded solution.

4 Integrated explanatory material

The following content develops the photographed material for study; it is not a claim of verbatim classroom speech:

1. Modeling as translating decisions into variables, objective and constraints, followed by testing and implementation.
2. Interior differentiable optima versus stationary points, separate treatment of endpoints and nondifferentiable points, and the limits of the second-derivative test.

3. A seven-step formulation method: scope, given parameters and units, decision variables and domains, objective, justified constraints, assumptions, validation and interpretation. This expands the six handwritten steps and the user’s explicit preference for clear modeling foundations.
4. The distinction between heuristics and methods with stated optimality or approximation guarantees.
5. Direct proof of the triangular LP optimum: $2x_1 + x_2 \leq 2(x_1 + x_2) \leq 2$, attained at (1,0).
6. Substitution checks and an algebraic optimality certificate for the cabinet model.
7. Continuous divisibility versus integer production, and definitions of general LP coefficients and index ranges.

No fabricated wood or labor units were added. The clean text uses generic resource units and requires unit consistency. The model maximizes revenue, not profit. The insurance comparison added in iteration 005 is now placed with the September 16 material, where both formulations have been introduced.

5 Mathematical verification

Triangular LP: maximize $2x_1 + x_2$ subject to $x_1 + x_2 \leq 1$ and nonnegativity. Vertices (1,0), (0,1), (0,0) have objective values 2, 1, 0. The optimum is (1,0).

Cabinet data, in product order:

Product	Wood	Labor	Revenue
Bookshelf	10	2	100
Cabinet with doors	12	4	150
Cabinet with drawers	25	8	200
Custom cabinet	20	12	400

Capacities are 5000 and 1500. At (375, 0, 0, 62.5), resource use is exactly 5000 and 1500, and revenue is 62500. Multiplying the wood constraint by 5 and the labor constraint by 25 gives

$$100x_1 + 160x_2 + 325x_3 + 400x_4 \leq 62500.$$

For nonnegative variables this bounds the objective above by 62500, attained by the stated solution. SciPy/HiGHS independently reproduced this solution in iteration 004; its result is retained in the batch’s `numerical_check.json`. No integer optimum is claimed.

6 Figure provenance

`figures/2026-09-14-feasible-region.pdf` is generated by `scripts/generate_figures.py`, with a PNG counterpart for Markdown. It reconstructs the triangle from IMG_7833:B01 and adds objective level lines and an improving-direction arrow. Coordinates, intercepts and objective value were checked. It is a mathematical reconstruction, not a reproduction of handwriting.

7 Administrative material kept outside the study PDF

IMG_7831:B01 contains HW1, September 21 before 4:00 pm, Overleaf, an AI-agent reminder, and both "6 problems" and "Expected to do 3/5". The problem-selection requirement is unresolved; these are personal reminders, not independently verified course announcements. An introductory abbreviation remains unreadable. They are preserved in the transcription and here rather than mixed with the theory.

8 Editorial and review status

The user explicitly requested clean definitive folder-based notes and a separate companion for sources, additions and corrections. This supersedes the previous inline labels. There are no known unresolved numerical readings in this day's study material. Review was performed by the assistant, not an independent human reviewer. The folder delivery command checks the document pair and records build hashes internally; compilation and page inspection are technical checks, not guarantees of infallibility.

September 16, 2026 — provenance and review

9 Scope and date evidence

Course: ISE 555, Advanced Deterministic Systems Optimization. Lecture date: September 16, 2026. IMG_7839 explicitly carries this date; the user confirmed that IMG_7835–IMG_7838 belong to the same date. All five originals remain in `raw/notes/14_09_2026/`; folder name does not override confirmed page dates. Confirmation is recorded in `work/2026-09-14/user_confirmations.json`.

The source transcription remains in `work/2026-09-14/transcription.md`. The clean study PDF contains correct statements and integrated explanations. This companion preserves the original mistakes and the reasons for their correction.

10 Section-to-source mapping

- Insurance workload allocation: IMG_7835:B01–B02, table and variable descriptions; IMG_7836:B01–B02, objective, worker load, assignment equalities and nonnegativity.
- Comparison of production and fraction-based modeling: editorial synthesis of the September 14 cabinet example and the September 16 insurance formulation.
- Variables, domains and feasibility: IMG_7837:B02–B03 and IMG_7838:B01.
- Optimality, global/local and strict minima: IMG_7838:B02–B03.
- Practical use of local solutions: IMG_7839:B01.

11 User-resolved readings

Worker 4, task P, was initially marked "15 or 19". The user explicitly confirmed **19** in iteration 005. The clean table and the corresponding load inequality use $19P_4 + 25Q_4 + 29R_4 \leq t$. The

prior ambiguity is retained only in historical records. Dates for IMG_7835–IMG_7838 are also resolved as September 16 by user confirmation.

12 Documented correction 1: equality represented by inequalities

IMG_7836:B02 writes $\sum P_i \leq 1$ and $\sum(-P_i) \leq 1$ as equivalent to $\sum P_i = 1$. The second right-hand side must be -1:

$$\sum P_i \leq 1, \quad -\sum P_i \leq -1.$$

The original pair permits all fractions to be zero, so it does not enforce full assignment. The study text teaches the correct pair and its interpretation only. The transformation $10P_1 + 28Q_1 + 31R_1 - t \leq 0$ was already correct.

13 Documented correction 2: binary domain

IMG_7837:B02 presents $x_i \in \{0, 1\}$ "or" $x \in \mathbb{Z}^N$. Integer membership alone does not enforce binary values. The study text states $x \in \{0, 1\}^N$, or equivalently integer components with $0 \leq x_i \leq 1$.

14 Documented correction 3: maximization and minimization

IMG_7838:B02 contains the boxed expression $\max f(x) = -\min f(x)$. The correct identity over the same feasible set is

$$\max_{x \in S} f(x) = -\min_{x \in S} [-f(x)].$$

The inner sign reversal is necessary. For example, on $[0,1]$, $\max x$ is 1 and $-\min x$ is 0, disproving the original identity. The clean notes use the correct identity and a positive example of the conversion, without repeating the mistake.

15 Documented correction 4: complete definitions of minima

IMG_7838:B03 describes strict minima by replacing $\leq$ with $<$. Strict comparisons must exclude $x = x^*$, otherwise they would require $f(x^*) < f(x^*)$. The final definitions explicitly require $x^* \in S$, and the local definition requires existence of $\epsilon > 0$. A strict local comparison uses $0 < \|x - x^*\| < \epsilon$. These conditions complete the abbreviated handwriting.

16 Integrated explanatory material and modeling assumptions

1. P, Q and R are treated as fractions of their respective complete workloads. A fraction of 0.3 means 30% of that task category, not 30% of the worker's total capacity.
2. Time coefficients multiply fractions proportionally. Workers execute their own assigned categories sequentially, while workers operate simultaneously. Under these assumptions, minimizing t minimizes the maximum worker load, not the sum of effort.

3. Four additional worker inequalities are expanded from the first worker's handwritten constraint using the table, including confirmed $P_4=19$.
4. The formulation has 15 allocation variables and t , five load inequalities, three full-assignment equalities and nonnegative fractions. Nonnegative loads imply $t \geq 0$.
5. The production-versus-fraction comparison and warning against imposing $P_i + Q_i + R_i = 1$ per worker are pedagogical additions requested to clarify modeling foundations.
6. General constraint signs are conventions: $g_i(x) \geq 0$ accommodates $Ax \leq b$ through $g_i = b_i - (Ax)_i$. Equality can be written as two opposite inequalities.
7. Practical acceptance of a local solution is distinguished from a certificate of global optimality.

The photograph's introductory insurance sentence is cut off. The word associated with P's workload is partly unclear; "information requests" follows the table heading. No time unit is specified. The study text uses a common abstract time unit and states workload/scheduling assumptions rather than inventing a fuller task statement. No numerical optimum for the insurance model is claimed or needed to explain formulation.

17 Numerical and symbolic checks

Worker	P coefficient	Q coefficient	R coefficient
1	10	28	31
2	15	22	42
3	13	18	35
4	19	25	29
5	17	23	33

All five constraints use these coefficients with a common upper bound t . Each category sums across five workers to 1. Variable domains and inequality signs were compared with the photographed equations; corrections are recorded above.

18 Figure provenance

figures/2026-09-16-workload-model.pdf, generated by `scripts/generate_figures.py`, shows three workload categories, five workers, full-assignment sums and load bounds. It is an explanatory schematic, not a computed allocation or a source photograph. The diagram's a_i, b_i, c_i mean the table's P/Q/R coefficients, including $a_4 = 19$. The inequality arrows, labels and structure were checked; a PNG counterpart is retained.

19 Source material kept outside the study PDF

IMG_7837:B01 includes an unclear assignment/Wednesday reminder and a unit outline: feasibility/constraints, optimality, convex functions/combinations and the incomplete heading "optimization by moving ...". These are preserved here and in the transcription. The unscaled feasibility sketch in IMG_7837:B02 has no boundary equations, so no numerical geometry was invented from it. The clean notes explain feasibility in words.

20 Review status

The user confirmed dates and a specific coefficient, and requested two definitive outputs per folder. This is not represented as independent human verification of every sentence. The assistant reviewed the mathematics and presentation. Residual issues are missing administrative/context text and unspecified units, recorded above. Neither dates nor the P4 coefficient remain open questions.